

Title: Effects of combined balance and plyometric training on athletic performance in female basketball players

Running title: Combined training in female basketball players

Laboratory

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ABSTRACT:

The purpose of this study was to examine the effect of 8 weeks combined balance and plyometric training on the physical fitness of female adolescent basketball players. Twenty six healthy regional-level players were randomly assigned to either an experimental group (E; $n = 16$, age = 16.4 ± 0.5) or a control group (C; $n = 10$, age = 16.5 ± 0.5). C maintained their normal basketball training schedule, whereas for 8 weeks E replaced a part of their standard regimen by biweekly combined training sessions. Testing before and after training included the Squat Jump (SJ), Countermovement Jump (CMJ), Drop Jump (DJ), 5, 10 and 20-m sprints, Stork Balance Test (SBT), Y-Balance Test (YBT) and Modified Illinois Change of Direction Test (MICODT). Results indicated no significant inter-group differences in SJ and CMJ height; however, E increased their DJ height ($p < 0.05$, Cohens' $d = 0.11$). No significant inter-group differences were found for sprint performance or SBT, but dynamic YBT tended to a significant group interaction ($p = 0.087$, $d = 0.006$). Post hoc analysis also showed a significant increase of MICODT for E ($\Delta 6.68\%$, $p = 0.041$, $d = 0.084$). In summary, the addition of 8 weeks of balance and plyometric training to regular in-season basketball training proved a safe and feasible intervention that enhanced DJ height, balance, and agility for female adolescent basketball players relative to the standard basketball training regimen.

Key Words: basketball; female athletes; plyometrics; balance training; jumping ability; sprinting; agility.

INTRODUCTION

Basketball is characterized by activities that range from low-intensity running and walking to maximal sprints and jumps. Some short sprints include rapid changes of direction and velocity, and agility is thus an important component of the skills required for this sport (13).

Previous authors have recommended both plyometric (21) and balance training (4, 18) to improve muscular strength and power, and to decrease the incidence and severity of sport-related injuries (14). Plyometrics is a dynamic form of resistance training with a rapid stretch-shortening cycle, involving both vertical and horizontal displacements of the individual's center of gravity (23). Component exercises include jumping, hopping, and skipping, with multi-joint actions, rapid eccentric phases, and high-velocity concentric muscular contractions potentiated by the stretch reflex (2, 18, 21). Such training is well suited to enhancing vertical jump height and running speed (10), rebound jump height (25), agility (18, 21), and balance (18).

Balance can be defined as the ability to maintain the body's center of gravity over its base of support; it reflects appropriate neuromuscular actions in response to continuous visual, vestibular and somato-sensory cues (5). A good balance is a prerequisite of success in almost every sport (3, 4, 9); it also enhances athletic performance (17, 22), and has value in the prevention of athletic injuries. Basketball players must address issues of physical contact and other situations challenging their balance, such as twisting movements (particularly of the pivot foot) during jump shots, offensive and defensive rebounds, accelerations and decelerations with changes of direction, penetrations into the defensive perimeter, boxing-out and, dribbling (5). These various actions are all performed in a limited space and require very fast movements, a high coordinating ability and appropriate strength (5).

Since balance and coordination are not fully developed in children of either sex (28), it is important that specific balance training be incorporated into programs for adolescents (3). A number of studies have shown improved balance-related test scores with balance training (16, 17, 22). Gandevia (15) further indicated that balance training could improve proprioceptive feedback, resulting in a more rapid and greater neuromuscular activation during plyometric exercise. Balance training may prepare for subsequent plyometric training by improving the mechanics of landing and take-off (20). Moreover, balance training may precondition a player for a subsequent resistance regimen (8).

Several authors have recently offered combined plyometric and balance training to young athletes (9, 16, 27). Zemková et al. (33) showed that 6 weeks of combined agility and balance training significantly enhanced dynamic balance, running speed, agility, and the ground contact time during a drop-jump relative to a group that received only agility training. An 8-week combined program of balance and plyometric exercises also proved superior to plyometrics alone in terms of improving sprint, and shuttle-run performance (9). However, to our knowledge there have been no prior studies of combined balance and plyometric in female adolescent basketball players.

This study thus aimed to examine the effects of replacing a part of the standard basketball training regimen by 8 weeks of combined balance + plyometric training in female adolescent basketball players. The initial hypothesis was that the experimental group would show greater gains of physical fitness than a control group who followed the standard basketball training regimen.

METHODS

Experimental approach to the problem

We examined whether 8 weeks of biweekly in-season combined balance and plyometric training would enhance selected aspects of performance in female U17 basketball players relative to their peers who continued to follow the standard in-season regimen. A group of 26 regional-level players (see Procedures and Evaluation for details of training) volunteered for random assignment to either an experimental group (E; $n = 16$) or a control group that continued to follow the standard in-season regimen (C; $n = 10$). Both groups had been training for 3 months, and were already 1 month into the competitive season. Data were collected before the modification of training, and after completion of the 8-week trial. The test protocol included assessments of jumping ability (Squat-jump, Countermovement-jump, Drop-jump), linear speed (5, 10 and 20-m sprints), change of direction test (Modified Illinois Change of Direction Test [MICODT]), and measures of balance (standing Stork Balance Test and Y-Balance Test). Initial and final testing was conducted at the same time of day (7:30-9:30 a.m.), and under the same experimental conditions, at least 48 hours after the last training session or match.

Subjects

Anthropometric data for E and C are presented in Table 1. Players who missed more than 20% of training sessions and/or more than two consecutive sessions were excluded from the study (26). All subjects were free of injuries during the six months prior to and throughout the study.

All procedures were in accordance with the latest version of the Declaration of Helsinki, and were approved by the University Institutional Review Committee for the ethical use of human subjects, according to current national laws and regulations and procedures. All participants and their legal representatives were first informed about testing and training procedures, as well as possible benefits and risks related to the study. Written informed assent from the children and their legal representatives was then obtained.

Insert table 1 about here

Procedures

Squat Jump, Countermovement Jump and Drop Jump

The heights of the Squat-jump (SJ), Countermovement-jump (CMJ) and Drop-jump (DJ) were determined using the Opto-jump photoelectric system (Microgate SRL Bolzano, Italy). Subjects began the SJ at a knee angle of 90 degrees, avoiding any downward movement, and performed a vertical jump by pushing upwards, keeping their legs straight throughout. The CMJ began from an upright position, with the subject making a rapid downward movement to a knee angle of ~90 degrees and simultaneously beginning to push-off. During the DJ, subjects were asked to rebound from a 0.3 m height box, immediately making a vertical jump. During all three jumps, subjects kept their hands on their hips and jumped as high as possible. One minute of rest was allowed between trials, with the highest jump on each test being used in subsequent analyses. The Optojump photoelectric cells consist of two parallel bars (a transmitter and a receiver unit), placed approximately 1 m apart. The transmitter contains 32 light-emitting diodes, positioned at 31-mm intervals, 3 mm above ground level.

Proprietary software (Opto-jump version 3.01.0001) connected to a personal computer allowed quantification of jump heights. The flight time of vertical jumps was measured with an accuracy of 1 msec (1 kHz). Jump height power relative to body mass (W/kg) was estimated as follows (7):

$$h = \frac{gt_f^2}{8}$$

$$Power = \frac{g^2 * t_f * t_c}{4 * t_t}$$

Where:

h: jump height

g: acceleration of gravity (9.81 m/s²)

t_f: flight time

t_c: contact time

t_t: total time of one jump = t_f + t_c

Intra-class correlation coefficients (ICC) were 0.91, 0.93, and 0.92 for SJ, CMJ, and DJ, respectively.

Running speed

Performance over a 20-meter sprint was assessed at 5-m, 10-m, and 20-m using an electronic timing system with 4 single beam photoelectric gates (Microgate, Bolzano, Italy). Subjects made a standing start 0.3-m before the first gate, which was placed 0.75-m above the ground to ensure it captured trunk movement and avoided false signals from limb motion. Three trials were separated by 6-8 min recovery intervals. The ICCs were 0.92, 0.94 and 0.97 over distances of 5 m, 10 m and 20 m, respectively.

Stork static balance test

The Stork test was performed on the dominant leg (27). Subjects stood with their opposite foot resting against the inside of the supporting knee and both hands on the hips. On the “go” signal, they raised the heel of their foot from the floor. The posture was held as long as possible, but the test was terminated when the heel of the supporting leg touched the ground or the foot moved away from the knee cap. The ICC was 0.90.

Y-balance test

Dynamic balance was assessed for the dominant leg using the lower quarter Y-balance test (27). Barefoot subjects maintain a single leg stance while reaching as far as possible with the contralateral leg in 3 directions (anterior, postero-medial and postero-lateral) set by fixing tape measures to the floor. The two posterior lines extended at an angle of 135° from the anterior line. Athletes’ leg lengths were first determined while lying supine, measuring from the anterior superior iliac spine to the most distal aspect of the medial malleolus. They then stood with the tip of their great toe at the center of the grid. They were asked to reach in the three directions while maintaining a single-limb stance. Trials were repeated if the participant (1) did not touch the required line with the reaching foot while maintaining weight bearing on the stance leg, (2) lifted the stance foot from the center of the grid, (3) lost balance at any point, (4) did not maintain start and return positions for one full second, or (5) touched the reaching foot to gain support. The maximal reach was measured in each direction. The composite score (CS) was calculated as: $CS = ([\text{maximum anterior reach distance} + \text{maximum postero-medial reach distance} + \text{maximum postero-lateral reach distance}] / [\text{leg length} \times 3] \times 100)$ (27). Three test trials were conducted in each direction, with two-minute rest intervals. Test-retest reliability for the different reach directions ranged between 0.90 and 0.95.

Modified Illinois change of direction test (MICODT)

This test was administered as outlined by Hachana et al. (19). Four markers defined a 5 m² area. Three central markers were placed 2.5 m apart. Participants started from a prone position, with the chin touching the starting line. They accelerated for 5 m, turned around, ran back to the starting line, and swerved in and out of 3 markers, completing 5-m sprints to finish the MICODT course. They were instructed to run around and not to cut over the markers; if they failed to follow these instructions, the trial was restarted after a 3 min. recovery period. The ICC was 0.94.

Details of training program

All subjects avoided any training other than that associated with the basketball team throughout the study. Every Tuesday and Thursday for 8 weeks, experimental subjects replaced a part of their regular training routine with the combined training program. A standardized 8-to15-minute warm-up was completed before every training session; this included low intensity running, coordination exercises, dynamic movements (lunges and skips), sprints, and dynamic stretching of the lower-limb muscles (26, 27). Combined training drills lasted around 45 minutes (30 min for balance exercises; 15min for plyometric). Sessions comprised 3 balance exercises, followed by 3 plyometric exercises. The number of repetitions, sets and the complexity of the exercises were increased progressively over the 8 weeks, to limit stress musculotendinous stress (Tables 2 and 3). Balance training was based on previous recommendations for intensity and volume (5). In the plyometric program, subjects were instructed to jump as high as possible in the vertical jumps, and as far as possible in the horizontal drills with a minimal ground contact time.

The total number of ground contacts per week increased progressively from 80 to 120 over the 8 weeks of training. Respective rest intervals of 90-s and a 30-s rest were allowed between each plyometric and balance exercise sets (27).

Insert Table 2 about here

Insert Table 3 about here

Statistical Analyses

Statistical analyses were carried out using the SPSS 20 program for Windows (SPSS, Inc., Chicago, IL). Normality of all variables was tested using the Kolmogorov-Smirnov procedure. Means and SDs were calculated, using standard statistical methods. Training related effects were assessed by 2-way mixed analyses of variance (group $\times$ time) with repeated measures on time. When group $\times$ time interactions reached statistical significance, post-hoc paired t-tests identified significant comparisons. Effect sizes (ESs) were determined by converting partial eta-squared to Cohen's d. According to Cohen (12) ESs can be classified as small ($0.00 < d < 0.49$), medium ($0.50 < d < 0.79$), and large ($d < 0.80$). Test-retest reliability was assessed using ICCs (12). The alpha level of significance was set at $p < 0.05$.

RESULTS

During the intervention period four control subjects dropped out of the trial because they left the training center for personal reasons. Baseline and post-intervention values for the remaining experimental and control subjects are summarized in (Table 4).

Insert Table 4 about here

Vertical-jump tests

There were no significant main effects of time for SJ ($p = 0.36$, $d = 0.018$) or CMJ ($p = 0.58$, $d = 0.006$), and no significant group $\times$ time interactions for SJ ($p = 0.19$, $d = 0.035$) or CMJ ($p = 0.14$, $d = 0.044$). However, DJ height showed a significant main effect of time ($p = 0.02$, $d = 0.09$) and a significant group $\times$ time interaction ($p < 0.05$, $d = 0.11$). Post-hoc analysis showed a significant increase in DJ height from pretest to posttest only in E ($\Delta 15.2\%$, $p < 0.001$, $d = 2.1$), but no significant main effect of time ($p = 0.519$, and $d = 0.19$) or group $\times$ time interaction ($p = 0.591$ and $d = 0.16$) in the measure of drop-jump power.

Running speed

No significant effects were found for any sprint interval ($p > 0.05$; $d = 0.068$, 0.063 and 0.064 for 5-m, 10-m, and 20-m, respectively), and no significant group $\times$ time interactions ($p = 0.87$, 0.32 , 0.75 ; $d = 0.001$, 0.02 , 0.002 , for 5-m, 10-m, and 20-m respectively)

Stork-balance test

A significant effect of time and a significant group $\times$ time interaction were observed for the SBT ($p = 0.002$, $d = 0.96$; $p = 0.01$, $d = 0.77$). Post-hoc analysis revealed a significant increase in SBT performance in E ($\Delta 127.17\%$, $d = 2.10$ and $p < 0.001$)

Y-balance test

The YBT showed a significant main effect of time ($p = 0.013$, $d = 0.123$), and a tendency toward a significant group $\times$ time interaction ($p = 0.087$, $d = 0.06$). Post-hoc analysis showed a significant increase of YBT performance in E ($\Delta 9.7\%$, $d = 3.92$ and $p < 0.001$).

Change of direction tests

A significant main effect of time and a significant group $\times$ time interaction were established for the MICODT ($p = 0.012$, $d = 0.124$; $p = 0.041$, $d = 0.084$). Post-hoc analysis showed a significant increase in MICODT performance in E ($\Delta 6.68\%$, $p < 0.001$, $d = 2.91$).

DISCUSSION

This study aimed to examine the effects on the physical fitness of female adolescent basketball players of replacing a part of standard basketball training by an 8-week combined balance + plyometric training. The main findings were of greater improvements in E than in C with respect to DJ, balance and MICODT performance. However the combined program did not achieve significantly greater gains in SJ, CMJ, or running speed.

There were no significant improvements in measures of SJ and CMJ. These results are in line with Zemková et al. (33), who found no significant improvements in SJ or CMJ after 6 weeks of combined agility and balance training in male basketball players (age = 20.9 ± 2.4 years). However, Roman et al. (24) observed a significant improvement in jumping ability (CMJ: [$\Delta 1.4\%$, $d = 1.3$]; SJ: [$\Delta 1.6\%$, $d = 0.6$]; DJ20: [$\Delta 1.2\%$, $d = 0.88$]; DJ40: [$\Delta 1.7\%$, $d = 0.68$]; and SLJ: [$\Delta 5.2\%$, $d = 0.16$]) after 10 weeks of contrast training in pre-puberal basketball players. Asadi et al. (1) also reported a significant increase in jumping ability in young basketball players (vertical jump: [$\Delta 14.1\%$, $d = 2.8$]; and broad jump: [$\Delta 4.8\%$, $d = 2.4$]) after 8 weeks of plyometric training. Further, Ramirez-Campillo et al. (29) showed a significant improvement in jumping ability (CMJ: [$\Delta 4.4\%$, $ES = 0.23$]; SJ: [$\Delta 5.1\%$, $ES = 0.27$]) after 6 weeks of plyometric training in female soccer players. These divergent results may reflect differences in the initial fitness level of study participants. Another factor may be the high number of jumps made during training sessions, relative to the requirements of the sport; during a basketball game, the percentage of jumps with the ball is $44.1 \pm 5.9\%$ and 15.8 ± 1.3 without the ball (13); this limits the potential for any significant improvement in E relative to C. Nevertheless, the present study showed a significant improvement in DJ height and power ($p = 0.016$; $p = 0.006$ respectively), extending the previous findings of Chelly et al. (11) who had observed a significant enhancement in drop jump height and power in addition to significant changes in SJ and CMJ performance of young male track athletes after 10-weeks of

plyometric training. Zemková et al. (33) also showed a reduction of ground contact time during the drop-jump [$p \leq 0.05$; from 0.294 ± 0.051 s to 0.261 ± 0.047 s], and an improvement in the ability to differentiate the force of muscle contraction during repeated jumps after a training program for basketball players.

The present findings indicated no change sprint times after 8 weeks of combined balance and plyometric training. This is in accordance with Thomas et al. (32), who saw no significant improvements of running speed in post-adolescent semi-professional soccer players (aged = 17.3 ± 0.4 years) after 6 weeks of either depth-jump or countermovement-jump training. Likewise, Hammami et al. (20) observed no significant improvements in the running speed of junior male soccer players (aged = 15.7 ± 0.2 years) after 8 weeks of plyometric training. In contrast, a recent study (24) saw a small but significant improvement in 25-m sprint times of pre-pubertal basketball players (mean age 8.7 ± 0.9 years) after 10 weeks of contrast training (Δ 0.3%, $d = 0.86$). Likewise, previous investigation (9) reported a significant improvement in 10 m sprint times of 12-15 years male adolescents (10m: Δ -5%, $d = -0.86$; 30m: Δ -2.8%, $d = -0.55$) after 8 weeks of a combined balance and plyometric training program. Asadi et al. (1) also showed a significant improvement in 60 m sprint performance of male basketball players after 8 weeks of plyometric training [Δ 16%, $d = -1.6$]. These discrepant findings probably reflect differences in methodology (male vs. female athletes, adolescent vs. much younger players; elite or professional players vs. regional level players; the format of the plyometric exercises, the frequency, duration, and progression of training; and its timing relative to the playing season). The optimal approach to enhancing sprint performance seems to incorporate greater horizontal acceleration (skipping, jumping with horizontal displacement) (31). An enhanced sprint performance is most likely caused by increases in neuromuscular activation of the trained muscles, including increases in the firing frequencies of active motor units and changes in their recruitment patterns (30).

Balance training studies have documented enhanced athletic performance (4, 17, 22), contributing to the prevention of athletic injuries and postural stabilization (6). The present findings showed a significant improvement ($p < 0.001$, $d = 2.1$) in static balance after 8 weeks of the combined training. Boccolini et al. (5) also found a significant increase in the sustained bi-podalic and mono-podalic positions ($p < 0.01$) of young male basketball players after 12 weeks of balance training. Likewise, Negra et al. (27) detected a significant improvement in performance of the stable stork ($\Delta 121\%$, $p < 0.01$, $d = 1.3$) and the unstable stork balance tests ($\Delta 149\%$, $p < 0.01$, $d = 1.83$) of prepubertal male soccer players after 8 weeks of unstable plyometric training. Others (9) reported a large increase in the Stork balance performance of adolescent male children (age = 13.3 ± 0.8 years) after 8 weeks of combined plyometric + balance training ($\Delta 87.8\%$, $d = 1.62$). The same study detected a smaller increase ($\Delta 59.6\%$, $d = 0.72$) in the Stork balance score after 8 weeks of plyometric training.

The present investigation showed a tendency toward significant interaction effects for the dynamic Y-balance test ($p = 0.087$, $d = 0.06$). In accord with the concept of training specificity (4), one would expect that a program that included balance activities would enhance balance scores relative to a program without such exercises. Others have shown enhanced dynamic balance, Zemková et al. (33) after agility-balance training, and Granacher et al. (18) a significant main effect of time for the star excursion balance score in sub-elite soccer players following 8 weeks of plyometric training on stable and unstable surfaces ($\Delta 2.8\%$, $p < 0.050$, $0.46 \leq F \leq 1.36$). Likewise, Negra et al. (27) observed improvements in the dynamic balance of pre-pubertal male soccer players (SYBT: $p > 0.05$, $d = 0.39$, UYBT: $p > 0.05$, $d = 0.08$), following plyometric training conducted on stable and unstable surfaces, and Granacher et al. (18) and Johnson et al. (21) both underlined that plyometric training can be safe and effective for youth.

The current study demonstrated a significant reduction of MICODT time after 8 weeks of the combined training program. This is consistent with Roman et al. (24) who noted enhanced change of direction performance in pre-pubertal male basketball players aged 8.72 ± 0.97 yr (Δ 1.6%, $d = 0.8$) after ten weeks of biweekly contrast training (isometric + plyometric) without external load. Likewise, Asadi et al. (1) found gains of agility (T-test [Δ -8.3%, $d = 2.2$] and Illinois change of direction test [Δ 5.8%, $d = 1.4$]) after 8 week of plyometric training program in young basketball players. Further, 8 weeks of combined plyometric +balance training yielded greater improvements in shuttle run times than plyometric training alone in 12-15 years old male adolescents (Δ -8%, $d = 0.52$) (9). Moreover, Ramiez-Campillo et al. (29) demonstrated increased change of direction speed performance (Δ 2.8%, $ES = 0.46$) after 6 week of plyometric training program in female soccer players, and Zemková et al. (33) showed a significant speeding of simple reaction time in an agility test ($\Delta = 5.8\%$) when a combined agility-balance training program on wobble boards was compared to agility training on a stable surface.

To the best of our knowledge the present study is the first to have investigated the effect of combined balance and plyometric on proxies of physical fitness in female adolescent basketball players. Our findings imply that female basketball players benefit from an in-season combined training program. Notably larger performance improvements were found in the experimental group compared to the control group in terms of SBT, agility, drop-jump performance, and the YBT performance.

Nevertheless, some methodological limitations warrant mention. Firstly the number of participants was limited ($n = 16$ for E. $n = 10$ for C respectively). Secondly, neuromuscular changes were not quantified, and ground reaction forces were not measured during jump and sprint performances.

Biomechanical and electrophysiological testing methods should be added to future studies to obtain in-depth knowledge regarding details of the adaptive processes after combined training.

PRACTICAL APPLICATIONS

The present study shows the feasibility of incorporating combined balance-plyometric training into a regular basketball training regimen for female U17 basketball players. A combined training program improves drop-jump height, dynamic balance and agility. On the other hand, such training is insufficient to improve squat-jump or countermovement jump height, or running speed. Therefore, the present findings should help coaches and exercise scientists in their quest to optimize daily training routines, and to enhance the performance of tasks critical to the success of female adolescent basketball players.

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Table1. Anthropometric characteristics of the study participants.

	E (n = 16)	C (n = 10)
Age (years)	16.4±0.5	16.5±0.5
Body height (m)	1.68±0.05	1.68±0.08
Body mass (kg)	56.6±8.3	55.6±7.0
Body mass index (kg/m ²)	19.9±2.1	19.8±2.7

Data are presented as means ± SD;

E: Experimental group; C: Control group

Table 2. Plyometric training program

Weeks	Ground contact per session	Plyometric drill	Sets x Rep
1	80	Vertical jump-and-reach Double leg jump forward 40cm hurdle jump	2 x 15 2 x 15 2 x 10
2	90	Vertical jump-and-reach Double leg zig-zag jump 40cm hurdle jump	2 x 15 2 x 15 2 x 15
3	90	Vertical jump-and-reach Double leg jump forward 60cm hurdle jump	2 x 15 2 x 15 2 x 15
4	99	Vertical jump-and-reach Double leg zig-zag jump 60cm hurdle jump	3 x 11 3 x 11 3 x 11
5	99	Vertical jump-and-reach Double leg jump forward 40cm drop-jump	3 x 11 3 x 11 3 x 11
6	108	Vertical jump-and-reach Double leg zig-zag jump 40cm drop-jump	3 x 12 3 x 12 3 x 12
7	108	Vertical jump-and-reach Double leg jump forward 60cm drop-jump	3 x 12 3 x 12 3 x 12
8	120	Vertical jump-and-reach Double leg zig-zag jump 60cm drop jump	3 x 13 3 x 13 3 x 14

Table 3. Balance training program

Balance drill	Sets x Rep	Rest interval (s)
Swiss-ball kneeling hold balancing	8 x 20 (s)	30
Two-handed chest pass balance exercise	6 x 20	30
Single leg balance	10 x 30 (s)	10

Table 4. Effect of eight-week combined balance-plyometric training on proxies of physical fitness in female basketball players.

	E (n=16)			C (n=10)			ANOVA		
	Pre	Post	$\Delta\%$	Pre	Post	$\Delta\%$	<i>p</i> -value (Cohen's <i>d</i>)		
	M $\pm$ SD	M $\pm$ SD		M $\pm$ SD	M $\pm$ SD		Group	Time	Group x Time
SJ (cm)	20.4 $\pm$ 3.9	22.5 $\pm$ 3.5	10.3	20.4 $\pm$ 2.5	20.0 $\pm$ 1.9	-1.81	0.19 (0.035)	0.36 (0.018)	0.19 (0.035)
CMJ (cm)	26.8 $\pm$ 3.8	28.8 $\pm$ 3.3	7.3	25.2 $\pm$ 2.9	24.4 $\pm$ 3.1	-3.52	0.003 (0.169)	0.583 (0.006)	0.144 (0.044)
DJ (cm)	24.7 $\pm$ 2.9	28.4 $\pm$ 3.0	15.2	24.8 $\pm$ 1.9	24.6 $\pm$ 2.8	-0.75	0.022 (0.105)	0.028 (0.096)	0.016 (0.115)
DJ (W/Kg)	26.5 $\pm$ 4.0	27.8 $\pm$ 5.0	5.2	23.9 $\pm$ 3.0	24.0 $\pm$ 3.2	0.52	0.008 (0.8)	0.519 (0.19)	0.591 (0.16)
5m sprint (s)	0.95 $\pm$ 0.08	0.91 $\pm$ 0.05	-4.14	1.01 $\pm$ 0.08	0.97 $\pm$ 0.06	-3.28	0.004 (0.164)	0.067 (0.068)	0.87 (0.001)
10m sprint (s)	1.82 $\pm$ 0.14	1.72 $\pm$ 0.09	-5.03	1.86 $\pm$ 0.11	1.83 $\pm$ 0.1	-1.40	0.025 (0.101)	0.078 (0.063)	0.32 (0.02)
20m sprint (s)	3.39 $\pm$ 0.25	3.27 $\pm$ 0.16	-3.17	3.46 $\pm$ 0.16	3.38 $\pm$ 0.16	-2.37	0.115 (0.051)	0.077 (0.064)	0.75 (0.002)
SBT (s)	17.3 $\pm$ 9.6	39.2 $\pm$ 13.3	127.2	21.8 $\pm$ 14.9	24.2 $\pm$ 14.7	11.01	0.162 (0.40)	0.002 (0.96)	0.01 (0.77)
YBT(s)	104.3 $\pm$ 9.7	114.4 $\pm$ 10.4	9.7	104.3 $\pm$ 4.0	106.0 $\pm$ 4.2	1.68	0.089 (0.059)	0.013 (0.123)	0.087 (0.06)
M ICODT (s)	11.3 $\pm$ 0.6	10.6 $\pm$ 0.4	-6.7	11.5 $\pm$ 0.6	11.5 $\pm$ 0.6	-0.71	0.001 (0.205)	0.012 (0.124)	0.041 (0.084)

Notes: M: mean; SD: standard deviation; $\Delta\%$: % improvement; DJ: Drop-jump; CMJ: Countermovement-jump; SJ: Squat-jump; SBT: Stork balance test; YBT: Y balance test; MICODT: Modified Illinois Change of Direction Test; d: Cohen's d (effect size).